COUNTRY PROFILES

Nuclear Power in Egypt

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- In the 1960s Egypt started up its first research reactor and began considering the use of nuclear electricity.
- Egypt is constructing its first nuclear power plant at El Dabaa, comprising four large Russian reactors. Once in operation, El Dabaa will supply 10% of the country's electricity.
- Russia will supply a dry storage facility for used fuel from the El Dabaa reactors.

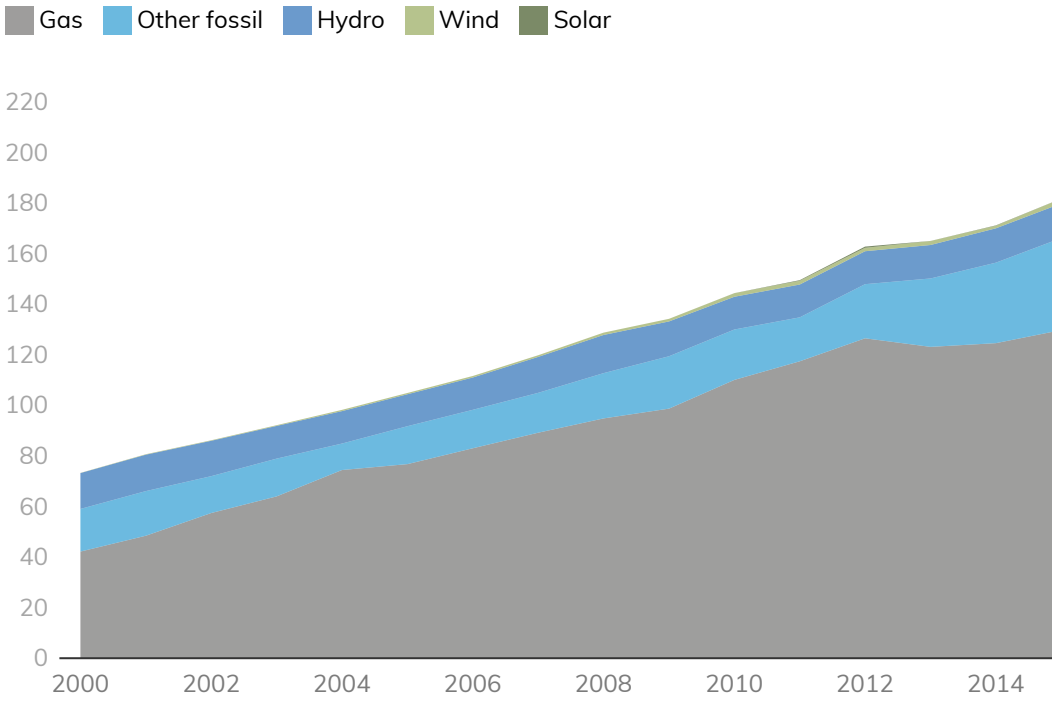
Privacy - Terms

El Dabaa will supply heat and power to an adjacent seawater desalination facility.

0	4	0
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
0 MWe	4,400 MWe	0 MWe

Electricity sector

Egypt electricity generation by source (TWh)



Total generation (in 2023): 215 TWh

Generation mix: natural gas 163 TWh (75.8%); oil 24.6 TWh (11.4%); wind 6.7 TWh (3.1%); hydro 15.5 TWh (7.2%); solar 5.3 TWh (2.5%).

Import/export balance: 1.4 TWh net export (0.1 TWh imports; 1.5 TWh exports)

Total consumption: 170 TWh

Per capita consumption: c. 1500 kWh in 2023

Source: International Energy Agency and The World Bank Data for year 2023.

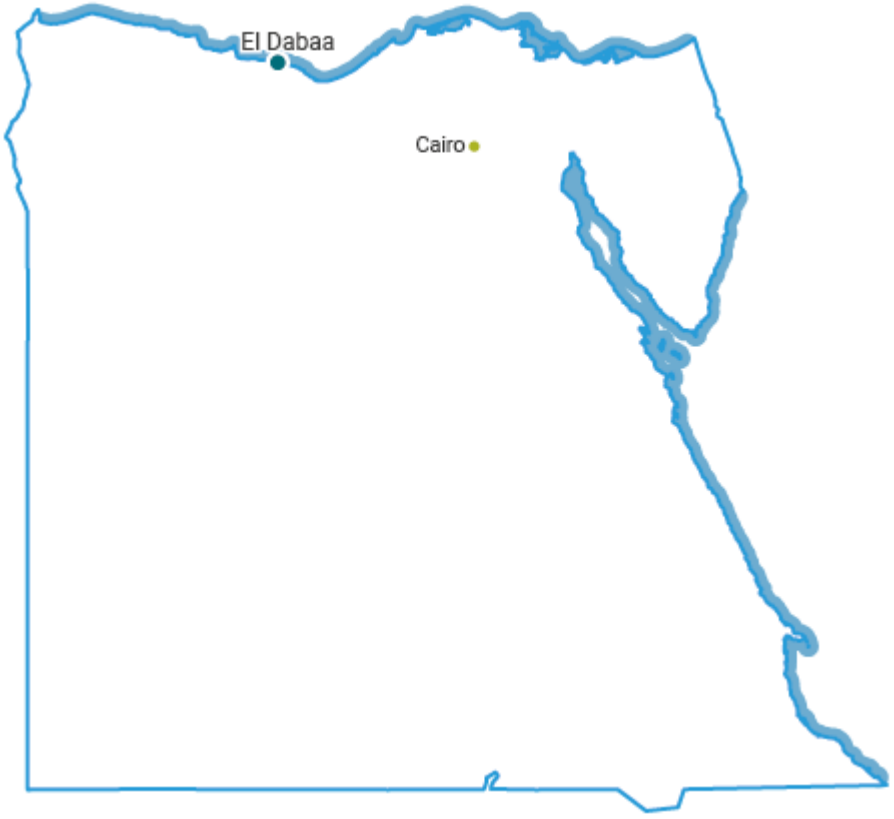
Energy demand growth in Egypt is significant. Per capita electricity consumption has increased from about 1000 kWh in 2000 to about 1500 kWh in 2022.

Egypt has long been reliant on natural gas for power generation. Annual domestic production of gas in 2023 was about 59.3 billion cubic metres (bcm), down about 11.5% year-on-year, and below consumption. Egypt became a net exporter of gas following the discovery of the giant Zohr gas field in 2015, but there have been few large discoveries since.

Nuclear power plans

Reactors under construction in Egypt

Reactor name	Model	Reactor type	Net capacity (MWe)	Construction start
El Dabaa 1	VVER-1200/V-529	PWR	1100	2022-07-20
El Dabaa 2	VVER-1200/V-529	PWR	1100	2022-11-19
El Dabaa 3	VVER-1200/V-529	PWR	1100	2023-05-03
El Dabaa 4	VVER-1200/V-529	PWR	1100	2024-01-23



El Dabaa

The El Dabaa site on the Mediterranean coast – 170 km west of Alexandria and Zafraana on the Gulf of Suez – was first selected to host a nuclear plant in 1983. Germany's KWU, Framatome and Westinghouse tendered to provide reactors for El Dabaa, and Australia and Niger agreed to supply uranium. The plan was aborted following the 1986 Chernobyl accident.

Over 1999-2001 the Nuclear Power Plants Authority (NPPA) carried out a feasibility study for a cogeneration plant for electricity and desalination, updating it in 2003. New nuclear cooperation agreements were signed with Russia in 2004 and 2008, reviving Egypt's plans for a nuclear power and desalination plant, supported by Rosatom. In October 2006 the Minister for Energy announced that a 1000 MWe reactor would be built at El Dabaa by 2015.

In December 2008, following an international tender, the Energy & Electricity Ministry awarded a $180 million contract to Bechtel to choose the reactor technology, choose the site for the plant, train operating personnel, and provide technical services over some ten years. However, in May 2009 the government transferred this contract to WorleyParsons, who signed it in June with the NPPA for $160 million over eight years to support the establishment of a 1200 MWe nuclear plant. The contract included site surveys and regional analysis to identify potential sites, comparing and ranking

them, and developing technical specifications for a planned tender. At the time the ministry said that Egypt aimed to begin generating nuclear electricity in 2017.

By early 2010 the proposal had expanded to four units by 2025. In March 2010 a legislative framework to regulate nuclear installations and activities in order to ensure the protection of facilities, individuals and property was signed into law. The first unit was expected to cost about $4 billion. In 2011 plans were put on hold until the country's political situation stabilized.

In April 2013 Egypt approached Russia to renew its nuclear cooperation agreement, focused on construction of a nuclear power plant at El Dabaa and joint development of uranium deposits. In October 2013 the Minister for Electricity & Energy reactivated plans for El Dabaa, and announced an NPPA site there. The Russian Foreign Minister said in November 2013 that Russia was ready to finance an Egyptian nuclear plant. In January 2014 the ministry said it would issue a tender, using WorleyParsons as consultants. In mid-2014 the target date for the tender was December 2014, and it was made clear that the winner would need to finance the plant. The tender would be for two units of 900 to 1650 MWe each on a turnkey basis. The El Dabaa site, situated close to El Alamein, is deemed suitable for eight reactors.

In February 2015 a further agreement was signed between Rosatom and the Ministry of Electricity and Renewable Energy, in which they "agreed to launch detailed discussions on the prospective project," involving construction of two 1200 MWe nuclear power units, with the prospect of two more. More specifically, Rusatom Overseas and the NPPA also signed a project development agreement for a two-unit AES-2006 nuclear power plant with desalination facility. Rosatom submitted a bid to build four 1200 MWe reactors at El Dabaa in June 2015.

In November 2015 an intergovernmental agreement was signed with Russia to build and operate the four reactors, including fuel supply, used fuel, training and development of regulatory infrastructure. The government also ordered preparations for construction of the units to begin in 2016. A financing agreement for a Russian state export loan was signed, covering 80% of the cost, with repayment over 22 years from 2029. In May 2016 the government announced that the loan was for $25 billion, to cover 85% of the cost of four 1200 MWe units, with repayments to start upon commissioning. In December 2017 notices to proceed with contracts for the construction of the four units were signed. Local content for the first unit is expected to be about 20%, increasing for

subsequent ones. The State Duma ratified the loan in November 2024.

Earlier in April 2019 the NPPA received a site approval permit for the El Dabaa site from the Egyptian Nuclear Regulation and Radiological Authority (ENRRA). In December 2019 the NPPA signed a further contract with Worley Ltd (formerly WorleyParsons) to serve as a consultant for the El Dabaa project to 2030, providing technical support for the NPPA's design review, project management, procurement, construction management, training, procedure development, quality assurance and commissioning of the plant.

The planned units at El Dabaa are designated as V-529, a warm-water version of the V-491 units at Leningrad II.

As well as addressing power supplies, the NPPA expects to build four nuclear desalination plants.

In November 2019 the IAEA concluded an integrated nuclear infrastructure review (INIR) undertaken at government invitation.

In August 2020 the NPPA said it expected a construction permit to be issued in the second half of 2021. In February 2021 representatives from the Russian and Egyptian governments reported that the Covid-19 pandemic had slowed preparations at the site, and by May 2021 the expectation was that a construction permit for unit 1 would be issued in July 2022. The NPPA applied to ENRRA for construction permits for units 1&2 in June 2021, and for units 3&4 in January 2022. In June 2022 ENRRA approved the construction permit for unit 1, and construction commenced in July. Construction for unit 2 began in November 2022, coinciding with Egypt's 'Nuclear Energy Day'.

In March 2023 ENRRA issued a construction licence for unit 3 and construction commenced at the beginning of May. ENRRA issued a construction licence for unit 4 in August 2023. First concrete for unit 4 was poured in January 2024.

In February 2025 Rosatom announced that new materials and welding technologies could increase the service lifetime of nuclear reactors to 100 years following the completion of welding for the VVER-1200 reactor pressure vessel (RPV) for El Dabaa 2.

In September 2025 the director of the construction project said 25,000 workers were active onsite. He expected operation of all four units by 2030.

In October 2025 the RPV for unit 1 was delivered after a 41-month manufacturing process in Russia. The RPV was installed in November.

Other proposals and agreements

Apart from Russia, nuclear cooperation agreements are in place with China (2006) and South Korea (2013). In May 2015 China National Nuclear Corporation (CNNC) signed an agreement with the NPPA to enhance nuclear cooperation and to "become an official partner" in the country's nuclear project.

In July 2015 it was reported that Korea Electric Power Co and Korea Hydro & Nuclear Power (KHNP) were making a combined bid to build 4000-6000 MWe in Egypt. Other proposals from China are reported. In November 2015 the government said the second set of four units would be put out to international tender.

In November 2016 the NPPA was working with WorleyParsons on a two-year project in the El Nagila area 80km east of Port Said to identify a suitable site for a second nuclear power plant.

Earlier in 1964 a 150 MWe nuclear plant with 20,000 m^3/day desalination capacity was proposed, then in 1974 a 600 MWe plant was proposed for Sidi Kreir near Alexandria. The government's NPPA was then established in 1976, and in 1978 plans were drawn up for ten reactors by 1999 with 7200 MWe capacity, at Sidi Kreir, Al Arish, Cairo and in Upper Egypt. Talks then with French, German and Austrian interests as well as Westinghouse came to nothing.

Fuel cycle

Rosatom will supply nuclear fuel for the entire lifecycle of the El Dabaa plant as well as supply containers for storing used fuel at the site.

In January 2025 the NPPA was granted a construction permit by the NRRA to construct a dry storage facility for used fuel at the El Dabaa plant. The facility will store used fuel for up to 100 years.

Research and development

Egypt has a 1961 vintage 2 MWt Russian research reactor at Inshas, serviced by Russia but currently in long-term shutdown; and a 22 MWt Argentinian research reactor (ETRR-2) partly supported by Russia that started up in 1997.

Egypt set up its own Atomic Energy Commission in 1955, which became the Egyptian Atomic Energy Authority (EAEA) the following year. It consists of three major centres – the Nuclear Research Centre (NRC), the National Centre for Radiation Research and Technology (NCRRT) and the Hot Laboratories and Waste Management Centre (HLWMC) – in addition to hosting the ETRR-2 and a cyclotron accelerator.

Regulatory, safety and non-proliferation

The Nuclear Regulatory and Safety Committee was established in 1982. As part of the reorganisation of the EAEA in 1991, this committee formed the National Center of Nuclear Safety and Radiation Control (NCNSRC) within the EAEA. In 2010, the Egyptian Nuclear and Radiological Regulatory Authority (ENRRA) was established as an independent nuclear regulatory body.

In 2015 a cooperation agreement was signed between Russia's Rostechnadzor and ENRRA.

Non-proliferation

Egypt signed the NPT in 1968, but until 1981 refused to ratify it unless Israel did. This caused plans in the 1970s to come to nothing. It has not signed the Additional Protocol.

Notes and references

General references

Y. Muralev, MAEK-Kazatomprom, Proposal of Desalination Technology Selection for Nuclear Power and Desalination Plant, presented at the International Atomic Energy Agency (IAEA) *Technical Meeting on the User-Vendor Interface in Cogeneration for Electricity Production and Seawater Desalination* held in Vienna, Austria on 14-16 March 2016

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